

Spec and Proj - Exercises

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These lecture notes from my website might be helpful.

1. A variety V is called *irreducible* if it can't be decomposed as $V = V_1 \cup V_2$ in a non-trivial way. Show that this is equivalent to (a) any two Zariski open subsets in V intersect, or (b) the ring $\mathbb{C}[V]$ is an integral domain.
2. (a) Show that Nullstellensatz fails over $\mathbb{R}$ by finding a polynomial f such that $I_{V(f)}$ is bigger than $\text{rad}(f)$. Generalize to any non-algebraically-closed field.
(b) Over $\mathbb{C}$ or $\mathbb{R}$ the ideal $I_{\mathbb{A}^n}$ is zero. What happens over $\mathbb{F}_p$?
3. (a) Show that for any hypersurface $V(f) \subset \mathbb{A}^n$, the complement $\mathbb{A}^n \setminus V(f)$ is isomorphic to an affine variety.
(b) Let $U \subset \mathbb{A}^2$ be the complement of the origin. Show that U is not isomorphic to any affine variety. *Hint: first show that the ring of regular functions on U is $\mathbb{C}[x, y]$ and deduce that any regular map from U to an affine variety must factor through the inclusion $U \hookrightarrow \mathbb{A}^2$.*
4. Prove that every homomorphism $\mathbb{C}[W] \rightarrow \mathbb{C}[V]$ is induced from a regular map $V \rightarrow W$.
5. Let R be the ring $R = \mathbb{C}[x]/(x^2)$. Show that a homomorphism $\mathbb{C}[V] \rightarrow R$ is exactly the data of a point $p \in V$ and a vector which is tangent to V at p .
6. Let $V = V(y^2 - x^3) \subset \mathbb{A}^2$ and let $f : \mathbb{A}^1 \rightarrow V$ be the function:

$$f : t \mapsto (t^2, t^3)$$

Show that f is not an isomorphism but it is a bijection (and even a homeomorphism in the usual complex topology).

7. Compute the transition functions (i.e. the co-ordinate changes) between the standard affine charts in $\mathbb{P}^2$.
8. Find an explicit method to compactify any hypersurface in $\mathbb{A}^n$ to a hypersurface in $\mathbb{P}^n$.
9. (a) Let C be the projective curve $V(xy - z^2) \subset \mathbb{P}^2$. Construct a bijection $\mathbb{P}^1 \rightarrow C$.
(b) You can interpret C as the space of non-zero 2×2 matrices with zero trace and determinant, up to scale. Find an abstract argument that this space is $\mathbb{P}^1$.
(c) What happens if $xy - z^2$ is replaced with another quadratic form?
10. (a) Consider the complex affine curve $V_\epsilon = V(xy - \epsilon) \subset \mathbb{A}^2$ for $\epsilon \in \mathbb{C}$. Convince yourself that (i) for $\epsilon \neq 0$ the space V_ϵ is a cylinder, and (ii) as $\epsilon \rightarrow 0$ one circle in the cylinder collapses, leaving two discs glued at a point.
(b) What's the topology of the complex projective plane curve $V(xy) \subset \mathbb{P}^2$? Now consider a small perturbation $f = xy + \epsilon g$ for some quadratic $g(x, y, z)$. What's the topology of $V(f)$? Compare this with Q9.
(c) Now take a cubic plane curve of the form $V = V(xyz + \epsilon h(x, y, z))$. Argue that V has the topology of a torus.